

Húsateikningaskoðun

CPU TIME LIMIT

2 seconds

MEMORY LIMIT

1024 MB

DIFFICULTY

medium (2.8)

The University of Iceland is planning to construct a new building that has a modern flair. The building should be on two floors and each floor should be a rectangle. The upper floor is wholly on top of the lower one. Your friend hands you a drawing of what he thinks the building will look like from above. Can you check if the image is correctly drawn?

Input

The first line of input contains two integers h and w ($1 \leq h, w \leq 50$). The next h lines each contain w characters. Each character is either '+' or '-', and both characters will each appear at least once. A '+' denotes the roof of the lower floor and '-' denotes the roof of the upper floor.

Output

If the input does not denote a possible image of the new building seen from above, print "Neibb". Otherwise print four integers h_1, w_1, h_2, w_2 where h_1 is the length of the lower floor in the image, w_1 is the width of the lower floor in the image, h_2 is the length of the upper floor in the image and w_2 is the width of the upper floor in the image.

Sample Input 1

```
3 4
++++
+-++
++++
```

Sample Output 1

```
3 4 1 1
```



Sample Input 2

```
5 7
+++++++
+- - - -+
+- - - -+
+++++++
+++++++
```

Sample Output 2

```
5 7 2 4
```

Sample Input 3

```
4 5
+++++
+- +++
++ - ++
+++++
```

Sample Output 3

```
Neibb
```